



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2002

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.

Total Marks - 84 marks

- All questions are of equal value.

Examiner: *E. Choy*

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject.

Question 1: [12 Marks]**Marks**

- (a) Evaluate $\int_{-2}^2 \frac{dx}{\sqrt{16-x^2}}$, giving your answer in exact form. 2

- (b) If $f(x) = e^{x+1}$ find the inverse function $f^{-1}(x)$ and hence show that 3

$$f[f^{-1}(x)] = f^{-1}[f(x)] = x$$

- (c) Solve the inequality 2

$$\frac{4-x}{x} \leq 1$$

- (d) Find the acute angle between the lines $y = \frac{1}{2}x$ and $x + \sqrt{3}y + 1 = 0$. 2
Give your answer in radians correct to two decimal places.

- (e) $A(10,1)$, $P(8,5)$ and B are points on the number plane. 3

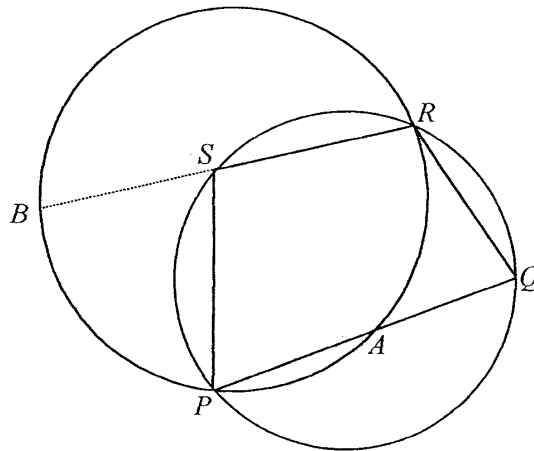
Point P divides the interval AB externally in the ratio 2: 3.

Find the coordinates of B .

Question 2: [12 Marks]

Marks

- (a) Differentiate $y = \tan^{-1}(\cot x)$ with respect to x . 2
- (b) Show that $\tan^{-1}(x) = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ 2
- (c) The polynomial $p(x) = ax^3 + bx^2 - 8x + 3$ has a factor $(x-1)$. When divided by $(x+2)$ the remainder is 15. 2
- Find the values of a and b .
- (d) Find $\frac{d}{dx}\left(\frac{\ln x}{x}\right)$ and hence find the primitive function of $\frac{2 - \ln x}{x^2}$ 2
- (e) The word EQUATION contains all five vowels. How many 3 letter “words” consisting of at least 1 vowel and 1 consonant can be made from the letters of EQUATION? 2
- [NB a “word” is ANY arrangement of the letters without any necessary meaning]
- (f) 2



$PQRS$ is a cyclic quadrilateral and A is any point on PQ .

A circle through the points P , A and R cuts RS produced at B .

Prove that $AB \parallel SQ$

Question 3: [12 Marks]**Marks**

- (a) Use mathematical induction to show that for all positive integers n

4

$$\sum_{r=1}^n a^{-r} = \frac{a^n - 1}{(a - 1)a^n}$$

- (b) The tangent at the point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the y -axis at T .

4

The line through the focus S parallel to this tangent cuts the directrix at V .

M is the midpoint of TV .

Find the locus of M as P moves on the parabola.

- (c) Show that $f(x) = x - 3 + \ln x$ has a root between $x = 1$ and $x = 3$.
If x_1 is this root, using Newton's method, prove that the second approximation is given by

4

$$x_2 = \frac{x_1(4 - \ln x_1)}{1 + x_1}$$

If $x_1 = 2$, find the value of x_2 giving your answer correct to two decimal places.

Question 4: [12 Marks]**Marks**

- (a) Tidal flow in a harbour is assumed to be simple harmonic motion and water depth x metres at time t hours is given by

$$x = 20 + A\cos(nt + \alpha)$$

where A , n and α are positive constants.

The depth of water is 12 m at low tide and 28 m at high tide which occurs 7 hours later.

- (i) Evaluate A and n . 3
- (ii) On a day when low tide occurs at 2.00 am, find the first time period during which the water level is greater than 22 m. 3

- (b) The acceleration of a body moving along a straight line is given by

$$\frac{d^2x}{dt^2} = -\frac{24}{x^2}$$

where x is the displacement from the origin after t seconds.

When $t = 0$, the body is 3 metres to the right of the origin with a velocity of 4 m/s.

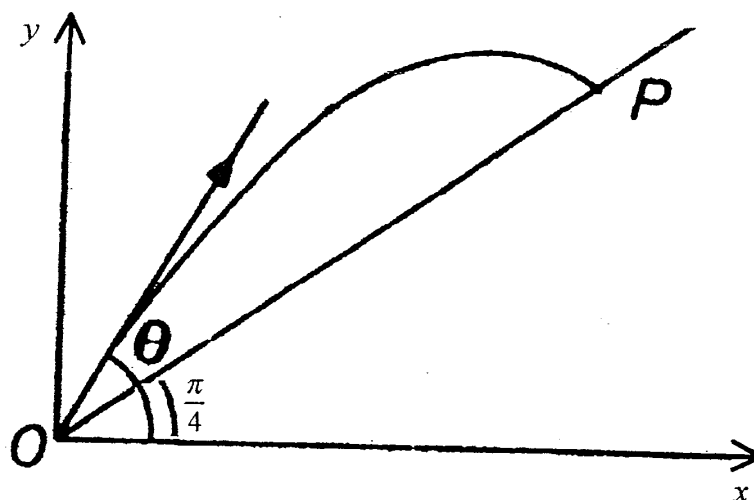
- (i) Show that the velocity, v , of the body in terms of x is given by 2

$$v = \frac{4\sqrt{3}}{\sqrt{x}}$$

- (ii) Find an expression for t in terms of x . 2
- (iii) How long does it take for the body to reach a point 10 m to the right of the origin? 2

Question 5: [12 Marks]

Marks



A golf ball is hit with a velocity of 5 m/s. It is projected at O , at the bottom of the slope inclined at $\frac{\pi}{4}$ to the horizontal.

The ball is projected at an angle θ to the horizontal, where $\frac{\pi}{4} < \theta < \frac{\pi}{2}$.

The equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -10$

- (i) Use calculus to show that the coordinates of the ball's position at time t seconds are given by 3

$$x = 5t \cos \theta \text{ and } y = -5t^2 + 5t \sin \theta$$

- (ii) The ball lands at P , where the length of $OP = R$ metres. 2

Show that $x = y = \frac{R}{\sqrt{2}}$

- (iii) Show that $R = 5\sqrt{2}(\cos \theta \sin \theta - \cos^2 \theta)$ 3

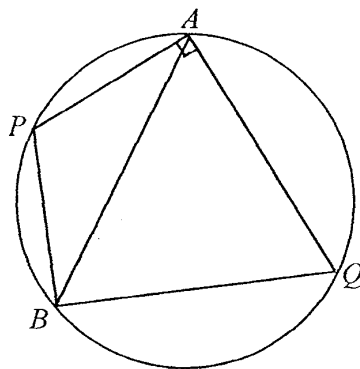
- (iv) By differentiation, find the exact value of θ (in radians) for the ball to achieve the maximum distance R . 2

- (v) Find the maximum value of R . 2

Question 6: [12 Marks]

Marks

(a)



A, P, B, Q are four points on a circle in a horizontal plane.

$$\angle AQB = \theta \text{ and } \angle PAQ = \frac{\pi}{2}$$

(i) Express $\sin \angle ABQ$ in terms of AB , AQ and θ

2

(ii) Hence find PQ in terms of AB and θ

3

(iii) Show that

2

$$PQ = \frac{\sqrt{AP^2 + BP^2 + 2AP \times BP \cos \theta}}{\sin \theta}$$

(b) (i) Prove that $\frac{\sin 2x}{1 - \cos 2x} = \cot x$

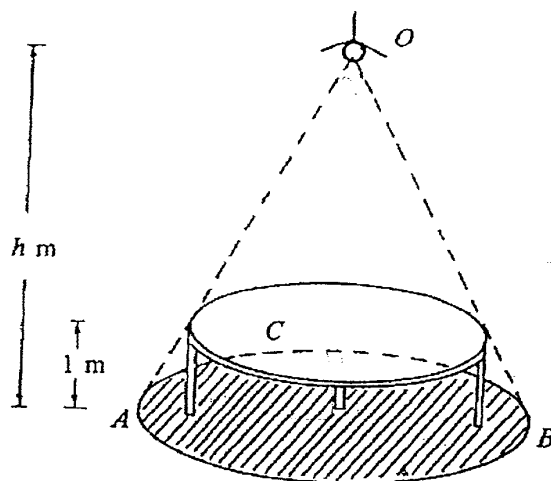
3

(ii) Hence, or otherwise, obtain a value for $\cot 67\frac{1}{2}^\circ$

2

Question 7: [12 Marks]

Marks



A small lamp O is placed h m above the ground, where $1 < h \leq 5$.

Vertically below the lamp is the centre of a round table of radius 2 m and height 1 m.

The lamp casts a shadow ABC of the table on the ground.

Let $S \text{ m}^2$ be the area of the shadow.

(i) Show that $S = \frac{4\pi h^2}{(h-1)^2}$ 3

(ii) If the lamp is lowered vertically at a constant rate of $\frac{1}{8} \text{ m/s}$, find the rate of change of S with respect to time when $h = 2$. 4

Let $V \text{ m}^3$ be the volume of the cone $OABC$.

(iii) Show that $V = \frac{4\pi h^3}{3(h-1)^2}$ 1

(iv) Find the minimum value of V as h varies. 4

Does S attain a minimum when V attains its minimum? Explain your answer.

THIS IS THE END OF THE EXAMINATION



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Mathematics Extension 1

Sample Solutions

Question 1

$$\begin{aligned} \text{x) } \int_{-2}^2 \frac{dx}{\sqrt{16-x^2}} &= \left[\sin^{-1} \frac{x}{4} \right]_{-2}^2 \quad (1) \\ &= \sin^{-1} \left(\frac{2}{4} \right) - \sin^{-1} \left(-\frac{2}{4} \right) \\ &= \frac{\pi}{6} + \frac{\pi}{6} \quad (1) \\ &= \frac{\pi}{3} \end{aligned}$$

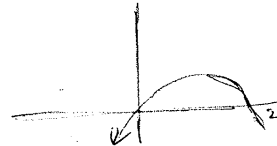
$$\begin{aligned} \text{(b) let } y &= e^{x+1} \\ \text{inverse } x &= e^{y+1} \\ \Rightarrow \log_e x &= y+1 \\ \text{ie } y &= \log_e x - 1 \quad (1) \\ \therefore f^{-1}(x) &= \log_e x - 1 \end{aligned}$$

$$\begin{aligned} \text{Now } f(x) &= e^{x+1} \\ \Rightarrow f[f^{-1}(x)] &= e^{f^{-1}(x)+1} \\ &= e^{\log_e x - 1 + 1} \\ &= e^{\log_e x} \quad (1) \\ &= x \end{aligned}$$

and

$$\begin{aligned} f^{-1}(x) &= \log_e x - 1 \\ \Rightarrow f^{-1}(f(x)) &= \log_e [f(x)] - 1 \quad (1) \\ &= \log_e [e^{x+1}] - 1 \\ &= (x+1) \log_e e - 1 \\ &= x+1 - 1 \\ &= x \end{aligned}$$

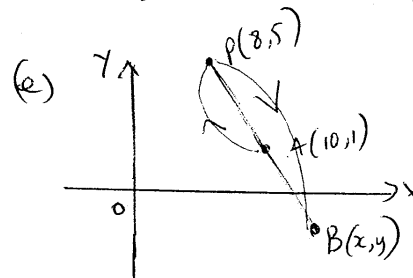
$$\begin{aligned} \text{(c) } \frac{4-x}{x} &\leq 1 \\ x(4-x) &\leq x^2 \\ 4x - 2x^2 &\leq 0 \quad (2) \\ 2x - x^2 &\leq 0 \end{aligned}$$



$$\therefore \text{ ~~xxxxxx~~ } \\ x \leq 0 \text{ or } x \geq 2$$

$$\begin{aligned} \text{(d) } \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{\frac{1}{2} - (-\frac{1}{\sqrt{3}})}{1 + \frac{1}{2}(-\frac{1}{\sqrt{3}})} \right| \quad (1) \\ &= \left| \frac{\sqrt{3} + 2}{2\sqrt{3} - 1} \right| \end{aligned}$$

$$\therefore \theta = \text{~~20.99~~ } 0.99^\circ \quad (1)$$



$$AP : PB = 2 : 3$$

$$A(x_1, y_1) \quad B(x_2, y_2) \quad m:n$$

$$\text{ie } A(10, 1) \quad B(x_2, y_2) \quad (1) \quad 2:3$$

$$\begin{aligned} \text{(1) } \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) &\equiv (8, 5) \quad (1) \\ \text{ie } \left(\frac{-2x_2 + 30}{1}, \frac{-2y_2 + 3}{1} \right) &\equiv (8, 5) \therefore B(-11, -1) \end{aligned}$$

Question 2

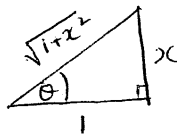
(a) $y = \tan^{-1}(\cot x)$

$$\frac{dy}{dx} = \frac{1}{1 + (\cot^2 x)} \cdot \frac{d(\cot x)}{dx}$$

$$= \frac{1}{1 + \cot^2 x} \cdot \frac{-1}{\sin^2 x}$$

$$= \frac{-\operatorname{cosec}^2 x}{1 + \cot^2 x} = \frac{-\operatorname{cosec}^2 x}{\operatorname{cosec}^2 x} = -1$$

(b) let $\tan^{-1} x = \theta \Rightarrow \tan \theta = x$



$$\sin \theta = \frac{x}{\sqrt{1+x^2}}$$

$$\therefore \theta = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \tan^{-1} x$$

(c) $P(x) = ax^3 + bx^2 - 8x + 3$

$$P(1) = a + b - 8 + 3 = 0$$

$$\therefore a + b = 5 \quad \text{--- (1)}$$

$$P(-2) = -8a + 4b + 16 + 3 = 15$$

$$\therefore -8a + 4b = -4 \quad \text{--- (2)}$$

$$\Rightarrow \underline{a = 2} \quad \underline{b = 3}$$

(d) $\frac{d}{dx}\left(\frac{\ln x}{x}\right) = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$

$$\therefore \frac{d}{dx}\left(\frac{\ln x}{x}\right) = \frac{1 - \ln x}{x^2}$$

$$\frac{d}{dx}\left(\frac{\ln x}{x}\right) + \frac{1}{x^2} = \frac{2 - \ln x}{x^2}$$

$$\therefore \frac{d}{dx}\left[\frac{\ln x}{x} - \frac{1}{x}\right] = \frac{2 - \ln x}{x^2}$$

$\therefore$ primitive of $\frac{2 - \ln x}{x^2}$ is

$$\frac{\ln x}{x} - \frac{1}{x} \text{ or } \frac{\ln x - 1}{x}$$

(e) ${}^5C_1 \times {}^3C_1 \times {}^6C_1 \times 3! = 540$

(f) $\angle SRP = \angle SQP$ (angles on same arc SP at circ.)

$\angle BRP = \angle BAP$ (angles on same arc BP at circ.)

$$\Rightarrow \angle SQP = \angle BAP \text{ which are corresp. angles formed by line}$$

SQ and BA, transversal QA
Since corresp. angles equal, lines SQ and BA must be parallel

Question 3

$$p(n): \sum_{r=1}^n a^{-r} = \frac{a^n - 1}{(a-1)a}$$

$p(1)$: Test for $n=1$

$$\begin{aligned} \text{LHS} &= \frac{1}{a} & \text{RHS} &= \frac{a-1}{(a-1)a} \\ & & &= \frac{1}{a} \\ & & &= \text{LHS} \end{aligned}$$

$\therefore p(1)$ is true

$p(k)$: Assume $p(n)$ is true when $n=k$ ($k \in \mathbb{N}^+$).

$$\therefore \sum_{r=1}^k a^{-r} = \frac{a^k - 1}{(a-1)a}$$

$p(k+1)$: Required to prove that $p(k) \rightarrow p(k+1)$.

$$\therefore \sum_{r=1}^{k+1} a^{-r} = \frac{a^{k+1} - 1}{(a-1)a^{k+1}}$$

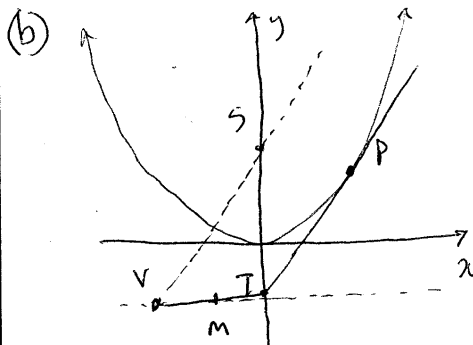
$$\begin{aligned} \text{LHS} &= \sum_{r=1}^k a^{-r} + a^{-(k+1)} \\ &= \frac{a^k - 1}{(a-1)a} + a^{-(k+1)} \\ & \quad \text{(by assumption)} \\ &= \frac{a(a^k - 1)}{(a-1)a^{k+1}} + \frac{1}{a^{k+1}} \\ &= \frac{a^{k+1} - a + (a-1)}{(a-1)a^{k+1}} \\ &= \frac{a^{k+1} - 1}{(a-1)a^{k+1}} \\ &= \text{RHS} \end{aligned}$$

$$\therefore p(k) \rightarrow p(k+1).$$

Since $p(1)$ is true,

$$p(1) \rightarrow p(2) \rightarrow p(3) \rightarrow \dots$$

By Principle of Mathematical induction, $p(n)$ is true for positive integral n .



$$\begin{aligned} \text{At } P(2ap, ap^2) \quad \frac{dy}{dx} &= \frac{dy}{dp} \cdot \frac{dp}{dx} \\ &= 2ap \cdot \frac{1}{2a} \\ &= p \end{aligned}$$

$$\therefore \text{Tgt at } P: y - ap^2 = p(x - 2ap)$$

$$\therefore px - y - ap^2 = 0$$

This line cuts y -axis when $x=0$

$$\therefore y = -ap^2$$

$$\therefore T \text{ is } (0, -ap^2)$$

The line thro' S || PT is

$$y - a = p(x - 0)$$

$$px - y + a = 0$$

This line cuts $y = -a$

$$px + a + a = 0$$

$$x = -\frac{2a}{p}$$

$$\therefore V \text{ is } \left(-\frac{2a}{p}, -a \right)$$

53/2

$$\therefore \text{For M: } x = \frac{-\frac{2a}{p} + 0}{2}$$

$$\therefore x = -\frac{a}{p} \quad \text{--- (1)}$$

$$y = \frac{-a + -ap^2}{2}$$

$$= -\frac{a(1+p^2)}{2} \quad \text{--- (2)}$$

For locus, eliminate p .

$$\text{From equation (1) } p = -\frac{a}{x}$$

$$\therefore y = -\frac{a\left(1 + \left(-\frac{a}{x}\right)^2\right)}{2}$$

$$2y = -a - a \times \frac{a^2}{x^2}$$

$$2y = -a\left(1 + \frac{a^2}{x^2}\right)$$

$$y = -\frac{a}{2}\left(1 + \frac{a^2}{x^2}\right)$$

$$(c) f(x) = x - 3 + \ln x$$

$$f(1) = 1 - 3 + 0$$

$$= -2$$

$$f(3) = 3 - 3 + \ln 3$$

$$= \ln 3$$

Since the sign of $f(x)$ changes between 1 and 3 and it is continuous in

the domain, there must be at least one root.

Newton's method state

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\text{Here } f'(x) = 1 + \frac{1}{x}$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{1 + \frac{1}{x_1}}$$

$$= x_1 - \frac{x_1 - 3 + \ln x_1}{1 + \frac{1}{x_1}}$$

$$= x_1 - \frac{x_1^2 - 3x_1 + x_1 \ln x_1}{1 + x_1}$$

$$= \frac{x_1(1+x_1) - x_1(x_1 - 3 + \ln x_1)}{1 + x_1}$$

$$= \frac{x_1(4 - \ln x_1)}{1 + x_1}$$

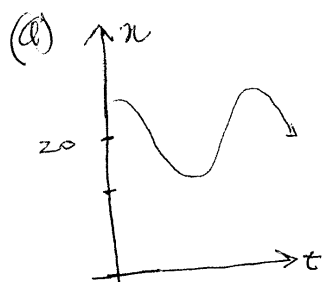
$$\text{Now if } x_1 = 2$$

$$x_2 = \frac{2(4 - \ln 2)}{1 + 2}$$

$$= \frac{8 - 2 \ln 2}{3}$$

$$\approx 2.20$$

Question 4



$$x = 20 + A \cos(\omega t + \alpha)$$

$$\text{Trough to crest} = 28 - 12 = 16$$

$$A = 8$$

$$\begin{aligned} \text{Period} &= 2 \times (\text{Trough to crest}) \\ &= 2 \times 16 \\ &= 32 \end{aligned}$$

$$\omega \quad 16 = \frac{2\pi}{\omega}$$

$$\omega = \frac{\pi}{7}$$

Let $t=0$ at 2:00 pm.
We seek.

$$22 = 20 + 8 \cos\left(\frac{\pi t}{7} + \alpha\right)$$

$$\text{Now } 12 = 20 + 8 \cos(0 + \alpha)$$

$$-8 = 8 \cos \alpha$$

$$\cos \alpha = -1$$

$$\alpha = \pi$$

$$\therefore 22 = 20 + 8 \cos\left(\frac{\pi t}{7} + \pi\right)$$

$$\frac{1}{4} = \cos\left(\frac{\pi}{7}t + \pi\right)$$

$$\frac{\pi}{7}t + \pi = \cos^{-1}\left(\frac{1}{4}\right) + 2k\pi$$

$$\frac{\pi}{7}t = \cos^{-1}\frac{1}{4} - \pi + 2k\pi$$

$$t = \frac{7}{\pi} \left(\cos^{-1}\frac{1}{4} + (k-1)\pi \right)$$

$$= 9.936 \text{ when } k=1$$

$$\therefore \text{Time } 2:00 \text{ pm} + 9 \text{ hr } 56' 13''$$

$$= 11:56:13 \text{ pm until}$$

$$= 16.936 \text{ when } k=2$$

$\therefore$ Tide remains above
22m from 11:56 pm till
4:56 am.

(b) $\frac{dv}{dx} = -\frac{24}{x^2}$

$$\text{When } t=0, x=3, v=4$$

$$(i) \quad \frac{d}{dx}\left(\frac{1}{2}v^2\right) = -\frac{24}{x^2}$$

$$\int \frac{d}{dx}\left(\frac{1}{2}v^2\right) dx = -24 \int \frac{dx}{x^2} + C$$

$$\frac{1}{2}v^2 = \frac{24}{x} + C$$

$$v^2 = \frac{48}{x} + C'$$

$$\text{When } x=3, v=4$$

$$16 = 16 + C'$$

$$C' = 0$$

$$\therefore v^2 = \frac{48}{x} \quad Q4/2)$$

$$\text{Now } v = \pm \frac{\sqrt{48}}{\sqrt{x}}$$

Since $v > 0$ initially,
we choose the positive
root.

$$\therefore v = \frac{4\sqrt{3}}{\sqrt{x}}$$

$$(i) \quad \frac{dx}{dt} = \frac{4\sqrt{3}}{\sqrt{x}}$$

$$\therefore \frac{dt}{dx} = \frac{\sqrt{x}}{4\sqrt{3}}$$

$$\text{So } \int \frac{dt}{dx} dx = \int \frac{\sqrt{x}}{4\sqrt{3}} dx + D$$

$$t = \frac{x^{3/2}}{\frac{3}{2} \times 4\sqrt{3}} + D$$

$$t = \frac{x\sqrt{x}}{6\sqrt{3}} + D$$

When $t=0$, $x=3$

$$0 = \frac{3\sqrt{3}}{6\sqrt{3}} + D$$

$$0 = \frac{1}{2} + D$$

$$\therefore D = -\frac{1}{2}$$

$$t = \frac{x\sqrt{x}}{6\sqrt{3}} - \frac{1}{2}$$

(ii) When $x=10$

$$\begin{aligned} t &= \frac{10\sqrt{10}}{6\sqrt{3}} - \frac{1}{2} \\ &= \underline{\underline{2.543 \text{ sec}}} \end{aligned}$$

(5)

(i)

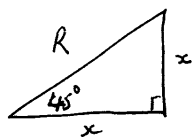
$$\begin{aligned}\ddot{x} &= 0 \\ \dot{x} &= c_1, \quad t=0, \dot{x}=5\omega\theta \\ \dot{x} &= 5\omega\theta \\ x &= 5t\omega\theta + c_2 \\ \boxed{x &= 5t\omega\theta} \\ c_2 &= 0 \text{ as } x=0 \text{ when } t=0\end{aligned}$$

$$\begin{aligned}\ddot{y} &= -10 \\ \dot{y} &= -10t + c_3 \quad \text{clearly } t=0, \dot{y}=5\omega\theta \\ y &= -10t + 5\omega\theta \\ y &= -5t^2 + 5t\omega\theta + c_4 \\ \text{clearly when } t=0, y=0 \\ \therefore \boxed{y &= -5t^2 + 5t\omega\theta}\end{aligned}$$

(3)

(2 marks)
for unit
velocity etc.

(ii)



$$\text{clearly } \frac{y}{x} = \tan 45^\circ = 1$$

$$\therefore y = x.$$

$$\text{and } \frac{x}{R} = \sin 45^\circ$$

$$x = \frac{R}{\sqrt{2}}.$$

$$\therefore \boxed{y = x = \frac{R}{\sqrt{2}}}$$

(2)

$$(iii) \text{ if } x = y, \quad 5t\omega\theta = -5t^2 + 5t\omega\theta$$

$$5t^2 + 5t\omega\theta - 5t\omega\theta = 0$$

$$5t(t + \omega\theta - \omega\theta) = 0$$

$$t = 0, \omega\theta - \omega\theta.$$

$$\therefore x = 5(\omega\theta - \omega\theta)\omega\theta$$

$$\therefore \boxed{R = 5\sqrt{2}(\omega\theta - \omega\theta)}$$

(3)

$$(iv) R' = 5\sqrt{2}[\omega\theta \times \omega\theta - \omega\theta \cdot \omega\theta - 2\omega\theta \cdot \omega\theta]$$

$$= 5\sqrt{2}[\omega^2\theta - \omega^2\theta + \omega^2\theta]$$

$$= 5\sqrt{2}[\omega^2\theta + \omega^2\theta]$$

$$\text{if } R' = 0$$

$$\omega^2\theta + \omega^2\theta = 0$$

$$\omega^2\theta = -\omega^2\theta$$

$$\tan 2\theta = -1.$$

$$R'' = 5\sqrt{2}[-2\omega^2\theta + 2\omega^2\theta]$$

$$= +10\sqrt{2}[\omega^2\theta - \omega^2\theta]$$

(2)

(1 1/2 if not
verified)

$$2\theta = \frac{3\pi}{4}$$

$$\theta = \frac{3\pi}{8}$$

$$R'' = +10\sqrt{2} \left[-\frac{1}{\sqrt{2}} \right]$$

$$= -20 \times 1.41$$

$$\therefore \text{MAX}$$

$$(v) \quad R = 5\sqrt{2} \left(\cos \frac{3\pi}{8} \sin \frac{3\pi}{8} - \cos^2 \frac{3\pi}{8} \right)$$

$$= 5\sqrt{2} \left(\frac{1}{2\sqrt{2}} - \left(\frac{1}{2} - \frac{1}{2\sqrt{2}} \right) \right)$$

$$= 5\sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right)$$

$$= 5\sqrt{2} \left(\frac{2-\sqrt{2}}{2\sqrt{2}} \right)$$

$$= \left| \frac{5}{2} (2-\sqrt{2}) \right| \quad \left(\begin{array}{l} \text{2 for exact} \\ \text{answer} \\ \text{1 for approx} \end{array} \right)$$

$$NB \quad \cos \frac{3\pi}{4} = 2 \cos^2 \frac{3\pi}{8} - 1$$

$$2 \cos^2 \frac{3\pi}{8} = 1 + \cos \frac{3\pi}{4}$$

$$\cos^2 \frac{3\pi}{8} = \frac{1}{2} + \frac{1}{2} \cos \frac{3\pi}{4}$$

$$= \frac{1}{2} + \frac{1}{2} \times -\frac{1}{\sqrt{2}}$$

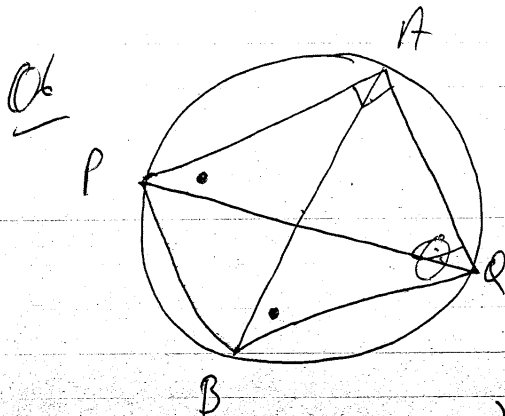
$$= \frac{1}{2} - \frac{1}{2\sqrt{2}}$$

$$\text{Also } \cos \frac{3\pi}{8} \sin \frac{3\pi}{8} = \frac{1}{2} \sin \frac{3\pi}{4}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}}$$

$$NB \quad \frac{5}{2} (2-\sqrt{2}) \doteq 1.464$$



$$i) \frac{AQ}{\sin \angle ABQ} = \frac{AB}{\sin \theta}$$

$$\therefore \sin \angle AQB = \frac{AQ}{AB} \sin \theta \quad (2)$$

$$ii) \text{ Now } \sin \angle AQA = \sin \angle AQB \quad (\text{equal angles in same circle})$$

$$\therefore \frac{AQ}{AB} \sin \theta = \frac{AQ}{PB}$$

$$\therefore \frac{\sin \theta}{AB} = \frac{1}{PB}$$

$$\therefore PB = \frac{AB}{\sin \theta}$$

$$(iii) \text{ Now } \angle APB = 180 - \cos \theta, \quad (\angle \text{ of cyclogram})$$

$$\text{In } \triangle APB: \quad AB^2 = BP^2 + PA^2 - 2BP \cdot PA \cos(180 - \theta)$$

$$AB^2 = AP^2 + BP^2 + 2AP \cdot BP \cos \theta$$

$$AB = \sqrt{AP^2 + BP^2 + 2AP \cdot BP \cos \theta} \quad (2)$$

$$\text{but } PB = \frac{AB}{\sin \theta} \quad \therefore AB = \frac{\sqrt{AP^2 + BP^2 + 2AP \cdot BP \cos \theta}}{\sin \theta}$$

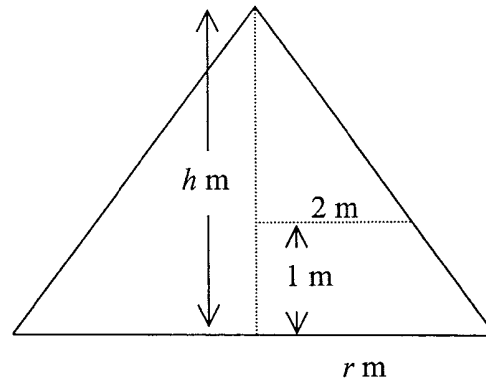
Q7 $i) \frac{\sin 2x}{1 - \cos 2x} = \cot x \Rightarrow 2 \sin x \cos x = \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)} = \frac{2 \sin x \cos x}{2 \sin^2 x}$

$$= \frac{\cos x}{\sin x} = \cot x \quad (3)$$

$$ii) \cot 67 \frac{1}{2}^\circ = \frac{\sin 135^\circ}{1 - \cos 135^\circ} = \frac{\frac{1}{\sqrt{2}}}{1 - (-\frac{1}{\sqrt{2}})} = \frac{\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} = \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{2} + 1}{\sqrt{2}}} = \frac{1}{\sqrt{2} + 1}$$

$$= \frac{1}{\sqrt{2} + 1} = \underline{\underline{\frac{1}{\sqrt{2} + 1}}} \quad (2)$$

(7)



(i)

$$S = \pi r^2$$

$$\frac{r}{h} = \frac{2}{h-1} \text{ (similar triangles)}$$

$$\therefore r = \frac{2h}{h-1}$$

$$\therefore S = \frac{4\pi h^2}{(h-1)^2}$$

(ii)

$$\frac{dS}{dt} = \frac{dS}{dh} \times \frac{dh}{dt}$$

$$S = \frac{4\pi h^2}{(h-1)^2}$$

$$\boxed{\frac{dh}{dt} = -\frac{1}{8}}$$

$$\frac{dS}{dh} = \frac{(8\pi h) \times (h-1)^2 - 4\pi h^2 \times 2(h-1)}{(h-1)^4}$$

$$= \frac{8\pi h(h-1)[(h-1) - h]}{(h-1)^4}$$

$$= -\frac{8\pi h}{(h-1)^3}$$

$$\frac{dS}{dt} = -\frac{8\pi h}{(h-1)^3} \times -\frac{1}{8} = \frac{\pi h}{(h-1)^2}$$

$$= 2\pi \text{ m}^2/\text{s when } h = 2$$

(iii)

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times \frac{4h^2}{(h-1)^2} \times h = \frac{4\pi h^3}{3(h-1)^2}$$

(iv)

$$\begin{aligned} V &= \frac{4\pi h^3}{3(h-1)^2} \\ \frac{dV}{dh} &= \frac{3(h-1)^2 \times 12\pi h^2 - 4\pi h^3 \times 6(h-1)}{9(h-1)^4} \\ &= \frac{12\pi h^2(h-1)[3(h-1) - 2h]}{9(h-1)^4} \\ &= \frac{4\pi h^2(h-3)}{9(h-1)^3} \end{aligned}$$

Minimum when $\frac{dV}{dh} = 0$

$$\frac{dV}{dh} = \frac{4\pi h^2(h-3)}{9(h-1)^3} = 0 \Rightarrow h = 0, 3$$

$$\therefore 1 < h \leq 5 \Rightarrow h = 3$$

h	2	3	4
$\frac{dV}{dh}$	-1	0	$\frac{1}{8}$

NB We only need

to test $\frac{(h-3)}{(h-1)^3}$

$$\therefore \frac{4\pi h^2}{9} > 0$$

So there is a *relative* minimum at $h = 3$

$$V = 9\pi$$

Testing end points $h = 5$, $V = \frac{125}{12}\pi$

So the minimum value of V is 9π , when $h = 3$

Note that $\frac{dS}{dh} \neq 0$ for $1 < h \leq 5$

So the minimum value of S will occur when $h = 5$, so the two minimums don't coincide for the same value of h .